# House Prices

## Advanced Regression Techniques

W207 Final Presentation

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# Competition Details

House Prices
Ames Housing dataset - Ames, Iowa

# Inference Problem

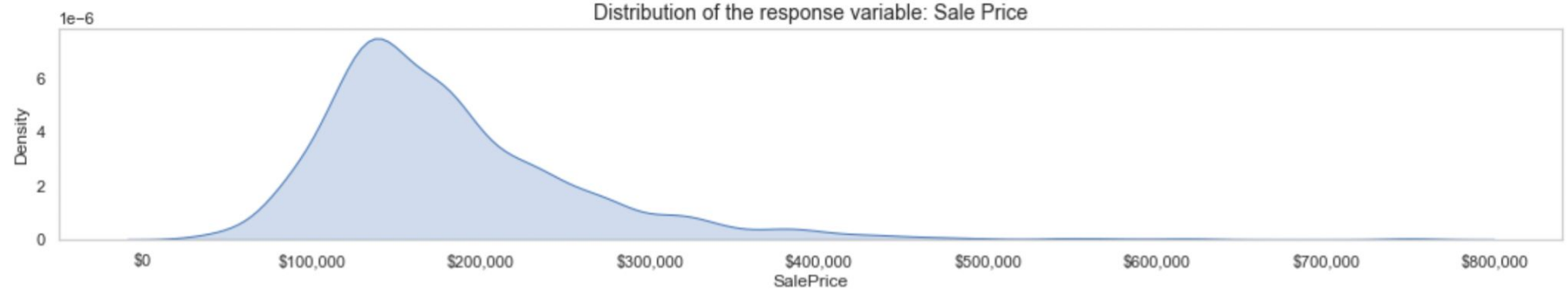
**X** ➡ **Y**

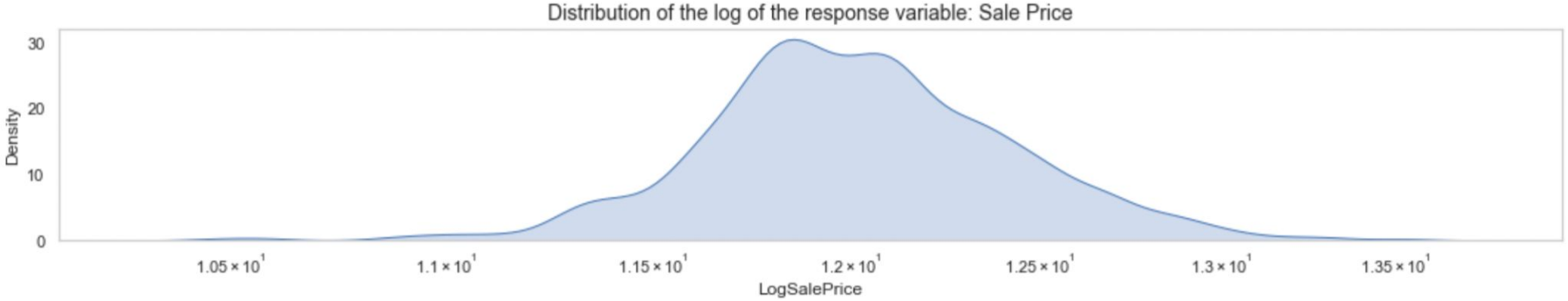
79 explanatory variables

Price of a house in Ames, Iowa

# EDA

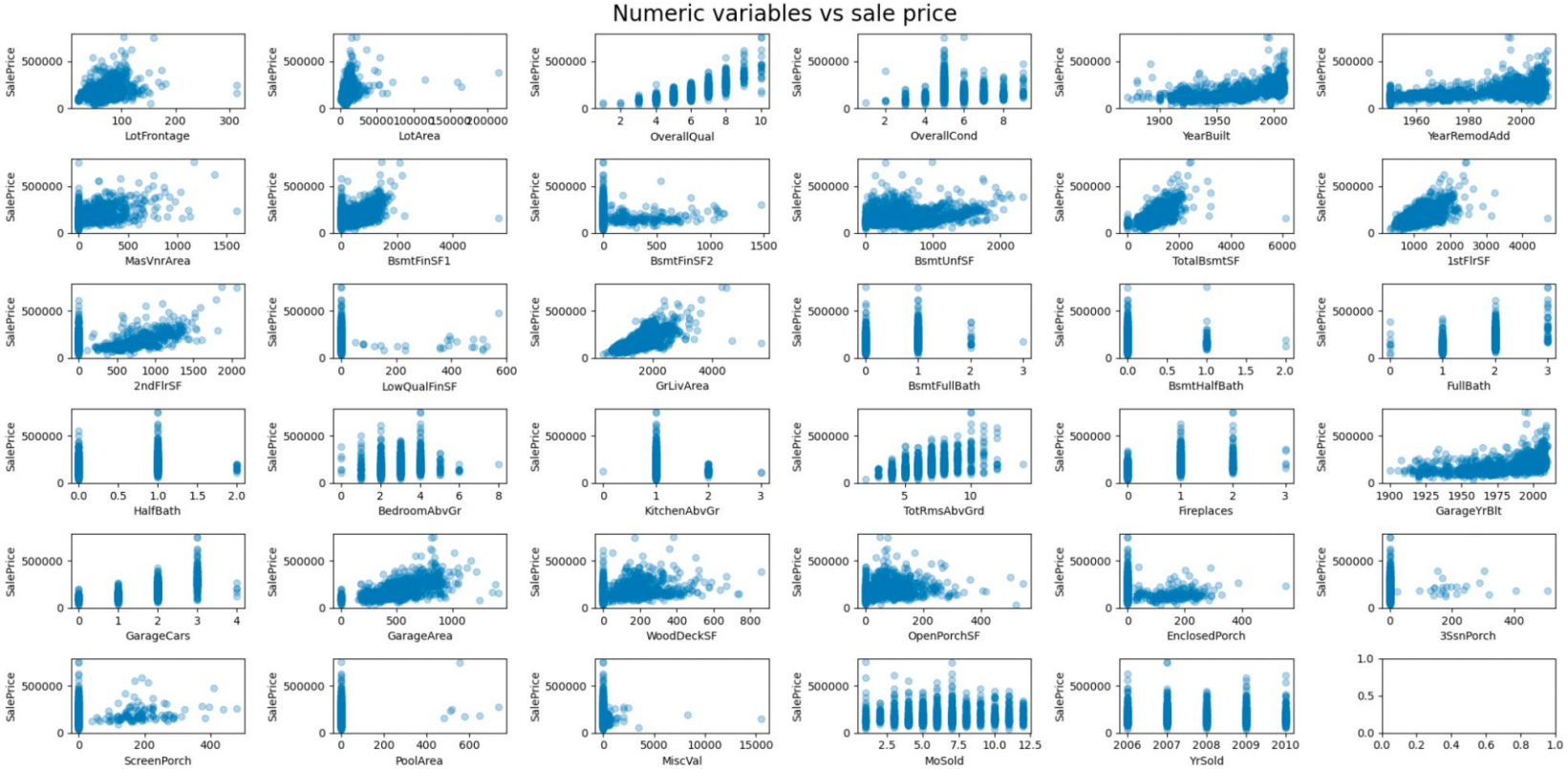
Sale Price



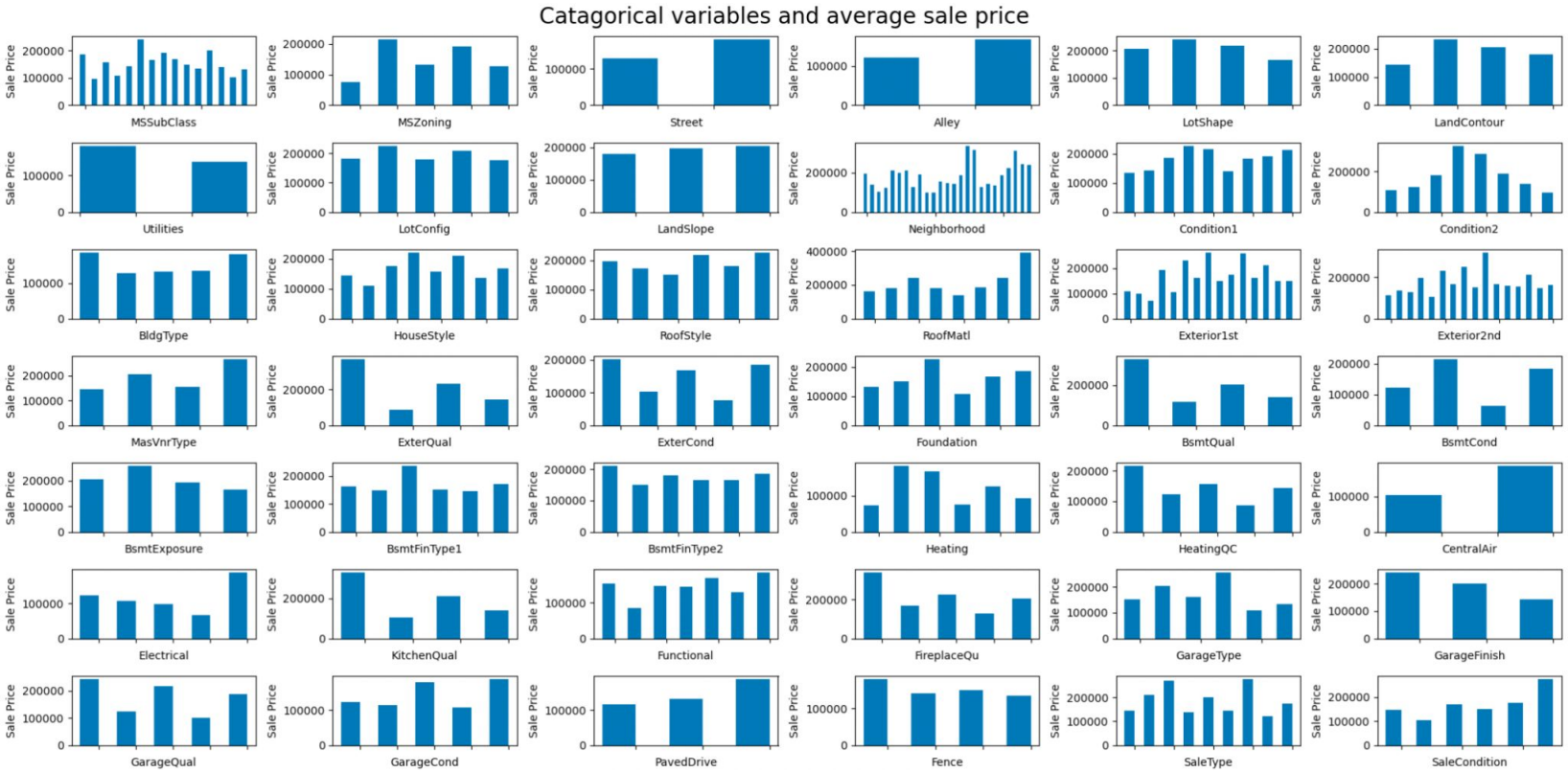
# EDA

Numeric Variables



# EDA

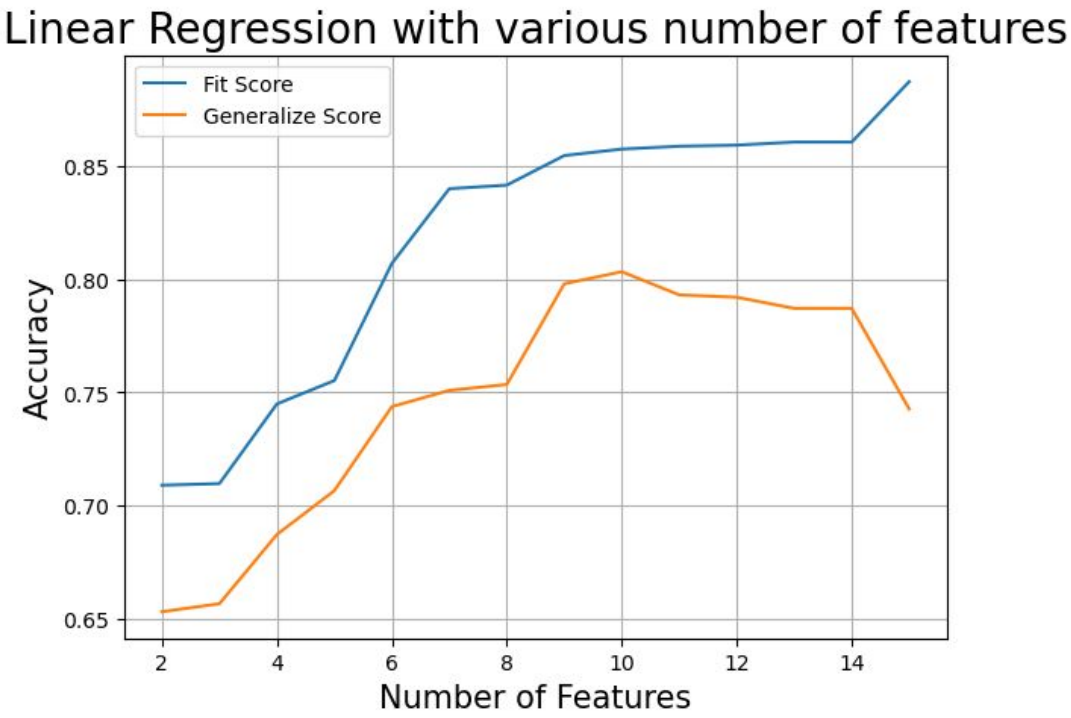
Categorical Variables



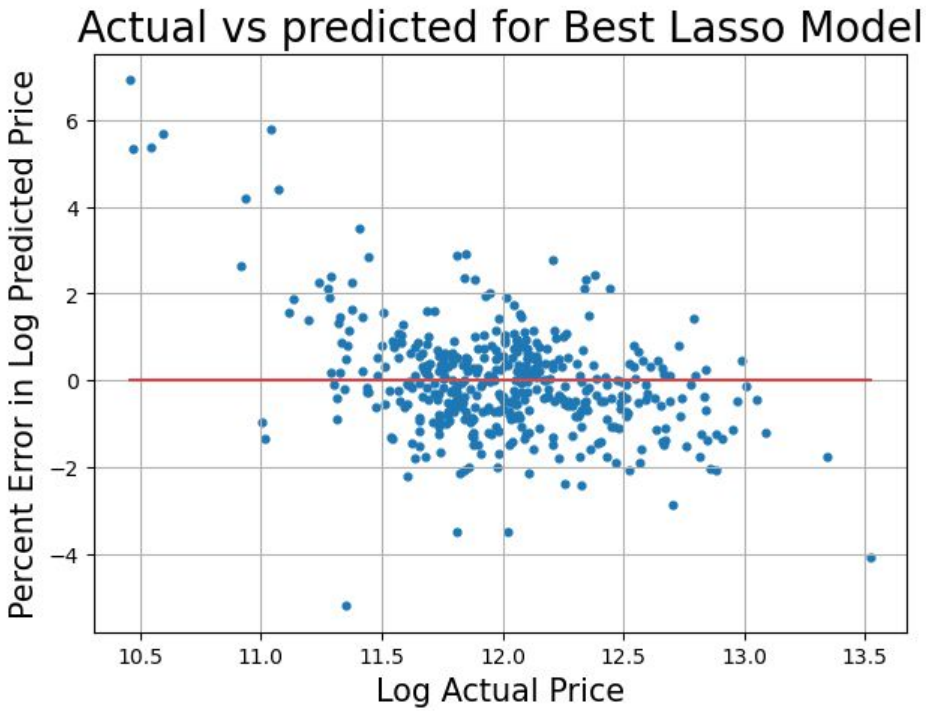
# Feature Engineering

- Combine Bathrooms into 1 variable
- Combine Sq feet area into 1 variable
- Add an indicator variable for whether a house is remodelled or not
- Log transform lot area, total inside area and basement sq footage (For Lasso)

# OLS Linear Regression: Complexity Analysis

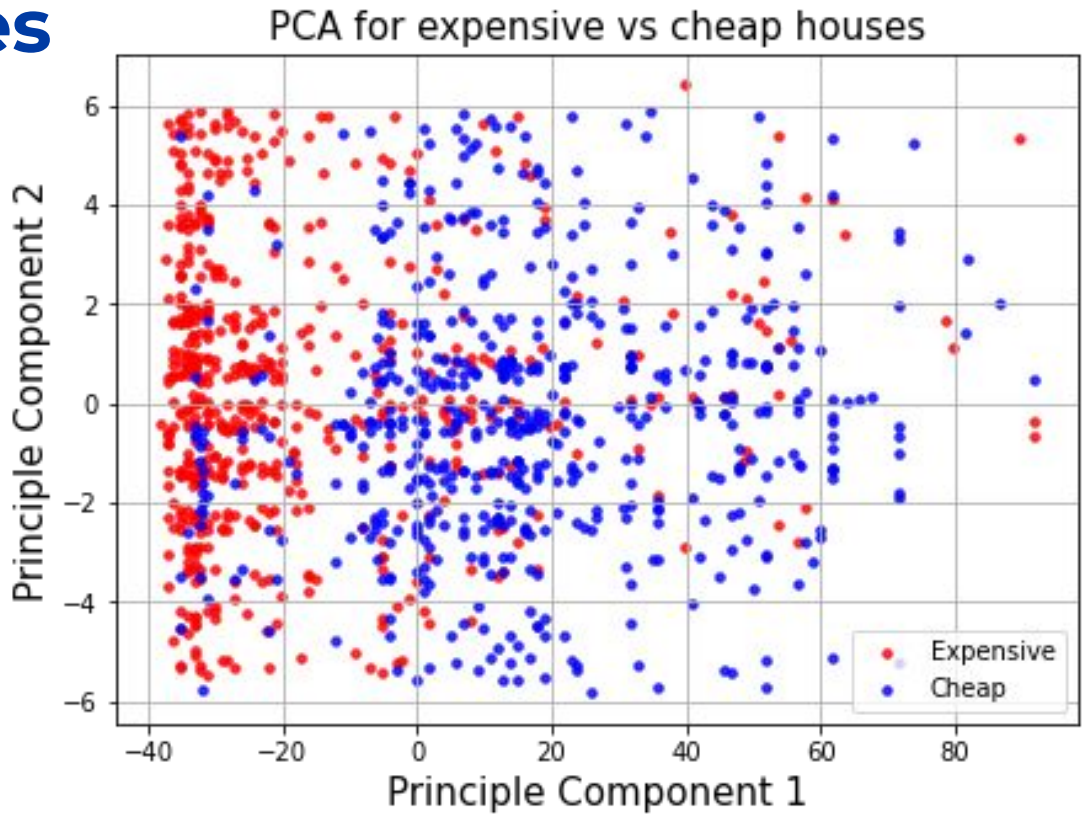


# Lasso Regression - Error Analysis




| OverallCond | YearBuilt | TotRmsAbvGrd | SalePrice | fraction_error |
|---|---|---|---|---|
| 5 | 1952 | 4 | 55993 | 0.041819 |
| 6 | 1927 | 5 | 64500 | 0.043958 |
| 3 | 1949 | 4 | 35311 | 0.053509 |
| 6 | 1910 | 6 | 37900 | 0.053613 |
| 4 | 1920 | 6 | 40000 | 0.056736 |
| 5 | 1965 | 5 | 62383 | 0.057706 |
| 5 | 1920 | 4 | 34900 | 0.069178 |

# PCA - Separation of Expensive/Cheap Houses



# Simple OLS

## Sale Price = $\beta_1$ x Overall Quality

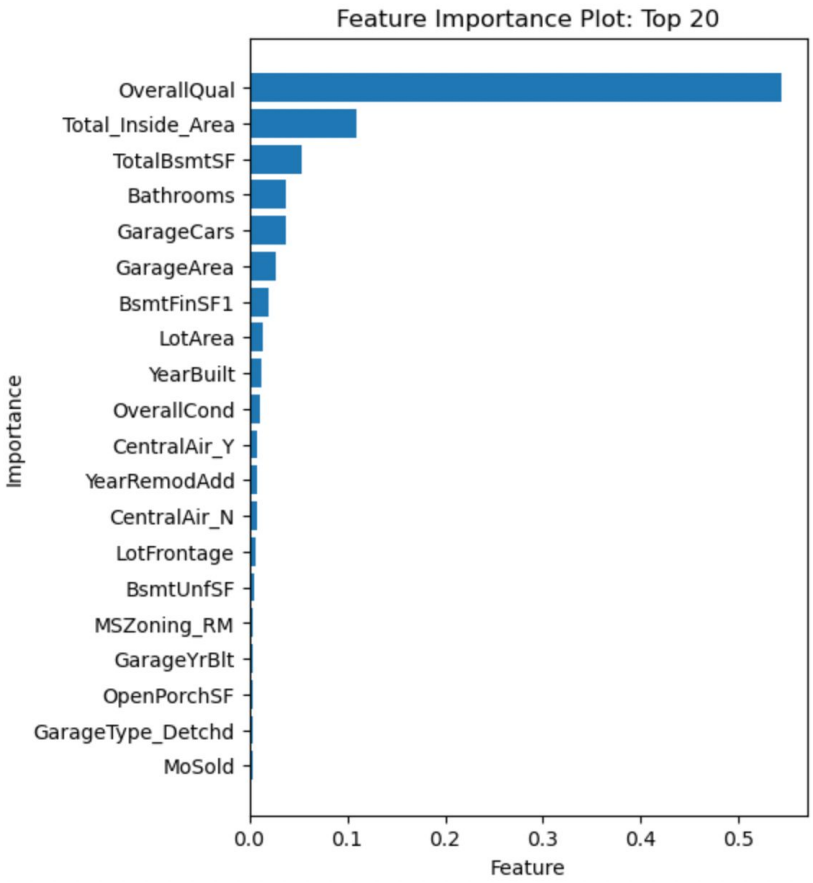
- Relatively low weight in final ensemble model
- Unknown exactly why its inclusion improves the fit of the overall model
- Potentially because overall model overfits and therefore including a largely underfit model improves the overall RMSE.

# Lasso Regression

- Relatively high weight in the final model
- Uses all numeric variables in the dataset after some data cleaning, and numericization of some categorical variables.
- Possibly performed well compared to OLS because of no collinearity assumptions and potentiality of weights to go to zero.
- Optimal alpha was very low.
- Assumes homoscedasticity, so four variables with large ranges of values were log-transformed

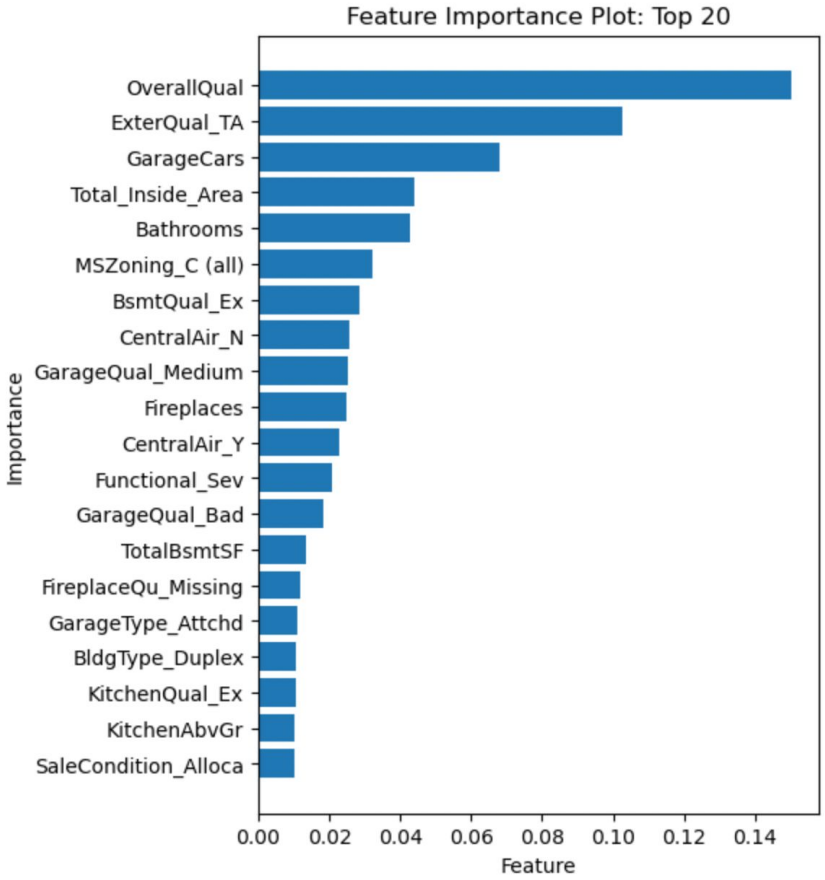
# Random Forest

- Moderate weight in the final model (30%)
- Includes all numeric and categorical variables including some feature engineered variables
- Feature importance plot shows features used most often in the splits
- Some important hyperparameters used:
  - Number of trees: 900
  - Maximum depth: 25
  - Maximum features: auto
  - Minimum samples for splitting: 2



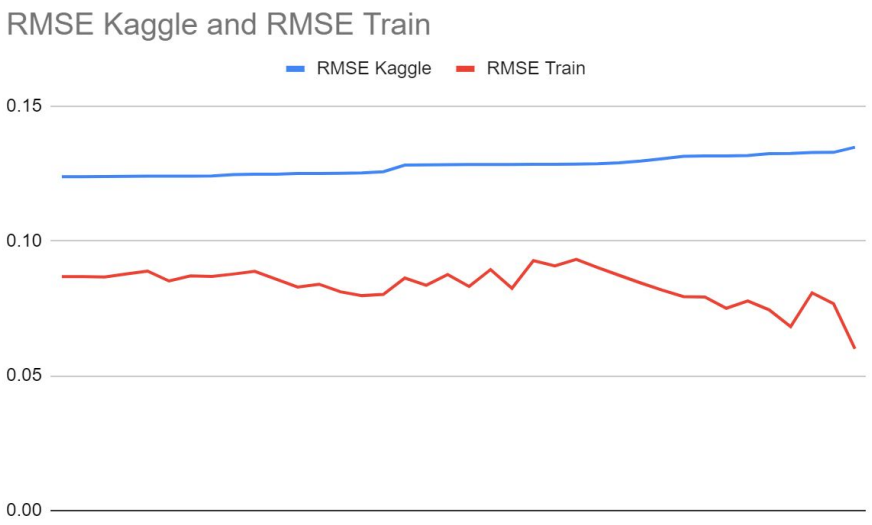
# XGBoost

- Low weight in the final model (5%)
- Includes all numeric and categorical variables including some feature engineered variables (like random forest)
- Feature importance plot shows features used most often in the splits
- Some important hyperparameters used:
  - Number of trees: 500
  - Maximum depth: 20
  - Maximum features: 70%
  - Learning rate: .05
  - L1 regularization: .01
  - L2 regularization: .1



# Final Model and Ensemble Weights

- **Final Weights**: Simple OLS (7.5%), Lasso Regression (57.5%), Random Forest (30%), XGBoost (5%)
- Weights difficult to optimize, over 50 configurations tested
- RMSE of training or testing data from our sample relatively uncorrelated with RMSE on Kaggle's unseen data.



# Summary

- **Final RMSE of 0.12298**
  - **Top 11% of Kaggle Submissions**

Highlights

- EDA and Feature Engineering
- Paths not taken
- Fine-tuning of final Ensemble Model

Additional Exploration

- Multiple models
- Explore RMSE discrepancy

# Questions?